Quadratic Kernel Mapping and Kernel Trick

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (u_1^2, u_2^2, \sqrt{2}u_1u_2) \cdot (v_1^2, v_2^2, \sqrt{2}v_1v_2) = u_1^2v_1^2 + u_2^2v_2^2 + 2u_1u_2v_1v_2$$

$$(\mathbf{u} \cdot \mathbf{v})^2 = (u_1v_1 + u_2v_2)^2 = u_1^2v_1^2 + 2u_1u_2v_1v_2 + u_2^2v_2^2$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (\mathbf{u} \cdot \mathbf{v})^2$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (3 \cdot 2 + 5 \cdot 1)^2 = 11^2 = 121$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (\sqrt{3}, 3\sqrt{2}, \sqrt{2}) \cdot (\sqrt{4}, 4\sqrt{2}, 2\sqrt{2}) = 3 \cdot 4 + 3\sqrt{2} \cdot 4\sqrt{2} + \sqrt{2} \cdot 2\sqrt{2} = 12 + 24 + 4 = 40$$

$$\mathbf{u} = (\sqrt{3}, 3\sqrt{2}, \sqrt{2}) \qquad \mathbf{v} = (\sqrt{4}, 4\sqrt{2}, 2\sqrt{2})$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (x_1^2, x_2^2, \sqrt{2}x_1x_2) \cdot (y_1^2, y_2^2, \sqrt{2}y_1y_2) = (\mathbf{x} \cdot \mathbf{y})^2$$

$$\text{where } \phi(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1x_2) \quad \rightarrow \quad (\mathbf{x} \cdot \mathbf{y})^2$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (u_1^2, u_2^2, \sqrt{2}u_1u_2) \cdot (v_1^2, v_2^2, \sqrt{2}v_1v_2) = u_1^2v_1^2 + u_2^2v_2^2 + 2u_1u_2v_1v_2$$

$$(\mathbf{u} \cdot \mathbf{v})^2 = (u_1v_1 + u_2v_2)^2 = u_1^2v_1^2 + 2u_1u_2v_1v_2 + u_2^2v_2^2$$

$$\phi(\mathbf{u}) \cdot \phi(\mathbf{v}) = (\mathbf{u} \cdot \mathbf{v})^2$$

$$\mathbf{u} = (u_1, u_2) \qquad \mathbf{v} = (v_1, v_2)$$